

the primary server crashes, the situation is detected by noticing missing live messages between the systems; and then redundancy hardware can take over the operations. The IP phones and other IP-based equipment can be easily switched over to the new server. A complicated case would involve switching over the physical layers of interfaces like FXO and FXS, which are directly connected to the server, for which physical switchover hardware is available.

Physical layer switchover hardware

Manufacturers provide physical hardware where, for example, the FXO line is connected. The output is duplicated and connected to the primary and secondary servers. The control signals to the device can come from the primary and secondary servers through the USB port or the IP port. As soon as a critical situation is discovered, the relays are activated and the physical connection is changed to the secondary server.

Paging systems

Paging systems could be overhead announcement speakers, which switch on, when dialled, or a group of phones which automatically switch on the speaker when dialled from a particular extension. Newer paging

systems also allow two-way communication, by allowing the receiver also to pass back messages.

SIP-based paging systems are available, which will accept the call as soon as the call comes, amplify the voice and output it through speakers. Asterisk supports creating multiple groups of paging devices and also incorporates the two-way messaging option.

Answering the front doorbell

A visitor might ring the house bell, and one can pick up an internal phone to communicate with him or her. There are a few SIP door phones available. Else, it is possible to connect an analogue phone with an FXO card to Asterisk.

We have now explored quite a lot of Asterisk hardware, which enhances its capabilities in the IP world by interfacing with other telecom networks, and carrying out multiple tasks. The power of applications on a simple switching platform is sometimes magical. 

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Figure 4: Exporting responses

ways. For example, if you need to distribute chocolates, you can ask people to choose between KitKat and Galaxy and based on the percentages, buy the right number of each brand.

Token management

In an organisation, one obviously doesn't want everyone to have access to every survey. For instance, the survey meant for the HR team may have nothing to do with the software developers in the firm. LimeSurvey provides an easy approach, called the 'token system', which limits the participation in surveys to only those who submit the proper

tokens. These tokens can be generated and distributed via e-mail to only the prospective group being surveyed.

Managing user groups

LimeSurvey can be set up on a server and can be accessed by multiple people simultaneously. This helps organisations to create a central hub for managing surveys. All users can access content and manage their surveys. This means that a technical manager can use the LimeSurvey for gathering a client's requirements, whereas the same server can be used by the HR department to deploy a survey asking employees their preferences on a particular subject.

Exporting and importing surveys

LimeSurvey focuses on re-usability and hence provides an easy way of exporting surveys. Exporting and importing surveys means saving a complete survey in a file and storing it, such that it can be used later. LimeSurvey exports surveys in the form of data-sets, which can be used when required.

LimeSurvey gives you complete flexibility in creating and managing surveys. Right from adding questions to releasing tokens, everything can be customised as per your requirements. Also, LimeSurvey allows you to add multiple themes by supporting plug-ins and survey themes. 

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